

Accessibility

Building User Interfaces

Cole Nelson

Make a copy of today's ICA and share it with your group members!

Learning Objectives

- Understand the relationship between accessibility and disability.
- Be able to apply the principles of accessible design, including specific applications to web design.
- Know the purpose and use of other assistive technologies.

What is accessibility?

Definitions

Usability: The effectiveness, efficiency, and satisfaction with which a specified set of users can achieve a specified set of tasks in a particular environment. — ISO 9241-11

Accessibility: The usability of a product, service, environment, or facility by people with the widest range of capabilities. — ISO 9241-20

From Accessibility to Disability

Accessibility is the extent to which an interactive product is accessible by as many people as possible.

The primary focus of accessible design is making systems accessible to individuals with *disabilities*.

Disability¹

Definition: A *disability* is any condition of the body or mind (impairment) that makes it more difficult for the person with the condition to do certain activities (activity limitation) and interact with the world around them (participation restrictions).

Disability can change over time with age or recovery, and the severity of the impact of disability can change over time. Fewer than 20% are born with a disability, although 80% of people will have a disability once they reach 85.

¹CDC

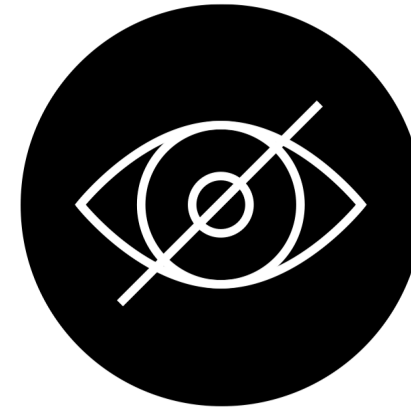
Three Dimensions of Disability²

1. **Impairment** in a person's body structure or function, or mental functioning (e.g., loss of a limb, loss of vision, or memory loss)
2. **Activity limitation** (e.g., difficulty seeing, hearing, walking, or problem solving)
3. **Participation Restrictions** in activities of daily living (e.g., working, engaging in social and recreational activities, and obtaining health care)

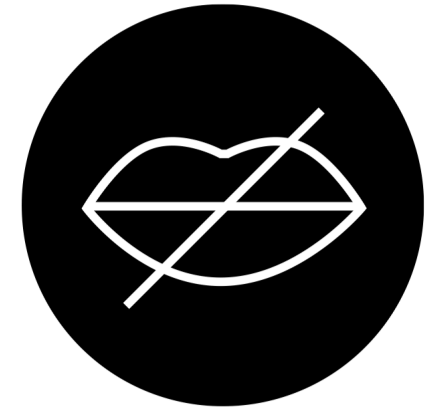
²Centers for Disease Control and Prevention (CDC)

Types of Impairment: Anatomical³

1. Sensory impairment
2. Physical impairment
3. Cognitive impairment



Can't see



Can't speak



Can't hear



Can't touch

³Image source: [Microsoft Inclusive Design Toolkit](#)

Sensory Impairment

Involves impairment in one or more senses, such as loss of vision or hearing.

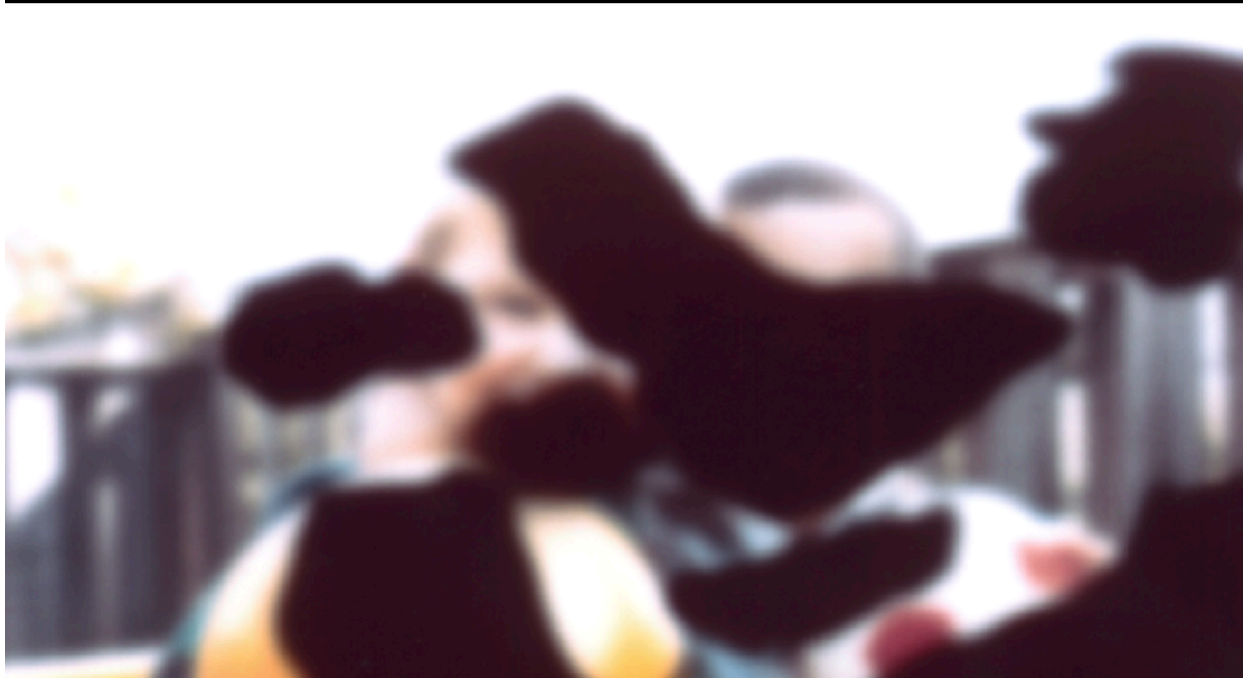
- Visual disabilities: Impairments in vision, including low vision, blindness, and color blindness.
- Auditory disabilities: Deficits that affect hearing at different levels of severity, including deafness.



Cataract



Age Related Macular Degeneration



Diabetic Retinopathy



Glaucoma

Physical Impairment

Involves loss of function to one or more parts of the body, e.g., congenitally or after stroke or spinal-cord injury.

- Motor/Mobility disability: Muscular or skeletal impairments in the hands or arms that affect user input as well as impairments that affect mobility, where users are in a wheelchair or bedridden.

Cognitive Impairment

Congenital, developmental, and traumatic conditions that result in cognitive deficits, such as learning impairment or loss of memory/cognitive function due to aging or conditions such as Alzheimer's disease.

Seizures

Definition: Neurological impairments, such as photosensitive epilepsy, that result in sensitivity to light, motion, and flickering on screen, which might trigger seizures.

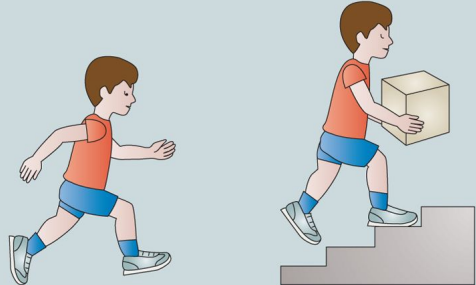
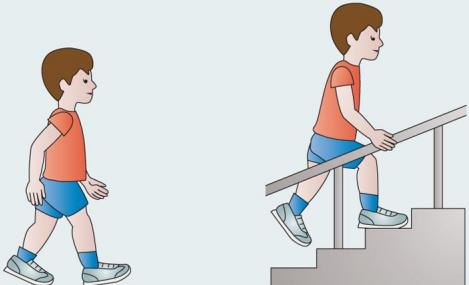
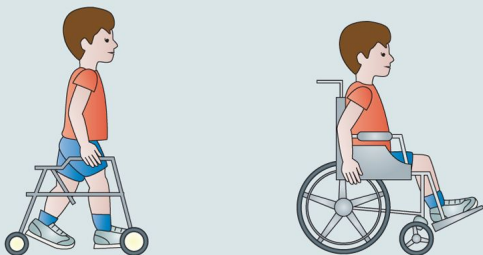
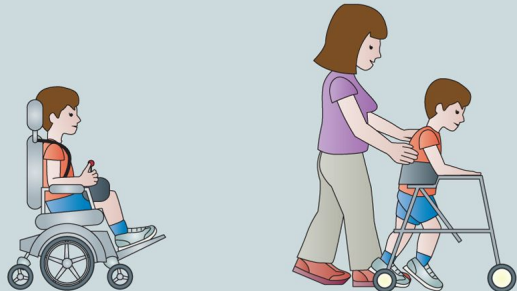
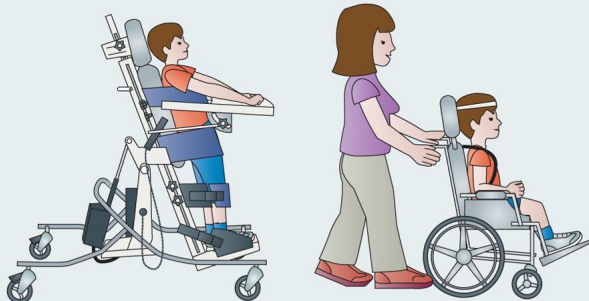
Variability⁴

Impairments can vary in severity or structure depending on the source and nature of the impairment.

Severity: Children with cerebral palsy can have basic mobility or completely depend on a caretaker.

Structure: Vision impairments can include color blindness, peripheral-only vision, no light perception

⁴ Image source

GMFCS expanded and revised between 6 th and 12 th birthday: descriptors and illustrations		
		GMFCS level I Children walk at home, school, outdoors and in the community. They can climb stairs without the use of a railing. Children perform gross motor skills such as running and jumping, but speed, balance and coordination are limited.
		GMFCS level II Children walk in most settings and climb stairs holding onto a railing. They may experience difficulty walking long distances and balancing on uneven terrain, inclines, in crowded areas or confined spaces. Children may walk with physical assistance, a hand-held mobility device or use wheeled mobility over long distances. Children have only minimal ability to perform gross motor skills such as running and jumping.
		GMFCS level III Children walk using a hand-held mobility device in most indoor settings. They may climb stairs holding onto a railing with supervision or assistance. Children use wheeled mobility when travelling long distances and may self-propel for shorter distances.
		GMFCS level IV Children use methods of mobility that require physical assistance or powered mobility in most settings. They may walk for short distances at home with physical assistance or use powered mobility or a body support walker when positioned. At school, outdoors and in the community children are transported in a manual wheelchair or use powered mobility.
		GMFCS level V Children are transported in a manual wheelchair in all settings. Children are limited in their ability to maintain antigravity head and trunk postures and control leg and arm movements.

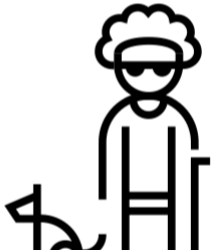
Types of Impairment: Temporal

1. *Permanent* impairment
2. *Temporary* impairment
3. *Situational* impairment

Permanent Impairment⁵

Congenital or long-term conditions, such as color blindness, missing body parts, etc.

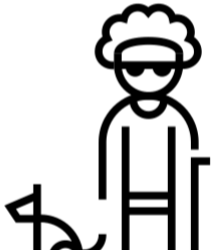
⁵Image source: [Microsoft Inclusive Design Toolkit](#)

	Permanent	Temporary	Situational
Touch	 One arm	 Arm injury	 New parent
See	 Blind	 Cataract	 Distracted driver
Hear	 Deaf	 Ear infection	 Bartender
Speak	 Non-verbal	 Laryngitis	 Heavy accent

Temporary Impairment⁶

Impairments that improve over time, such as recovery after illness or accidents, e.g., a broken arm.

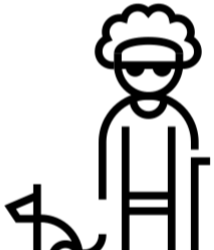
⁶Image source: [Microsoft Inclusive Design Toolkit](#)

	Permanent	Temporary	Situational
Touch	 One arm	 Arm injury	 New parent
See	 Blind	 Cataract	 Distracted driver
Hear	 Deaf	 Ear infection	 Bartender
Speak	 Non-verbal	 Laryngitis	 Heavy accent

Situational Impairment⁷

Impairments introduced by context, such as environments with low light or noise.

⁷Image source: [Microsoft Inclusive Design Toolkit](#)

	Permanent	Temporary	Situational
Touch	 One arm	 Arm injury	 New parent
See	 Blind	 Cataract	 Distracted driver
Hear	 Deaf	 Ear infection	 Bartender
Speak	 Non-verbal	 Laryngitis	 Heavy accent

How do we achieve accessibility?

Two ways to address accessibility problems:

1. Accessible design
2. Assistive technologies

Accessible Design

Medical Model of Disability

Disability as **personal attribute**

In the context of health experience, a disability is any restriction or lack of ability (resulting from an impairment) to perform an activity in the manner or within the range considered normal for a human being.

Social Model of Disability

Disability as **context dependent**

Disability is not just a health problem. It is a complex phenomenon, reflecting the interaction between features of a person's body and features of the society in which he or she lives.

Social Model ⁸:

People are disabled by barriers in society, not by their impairment.

⁸Shakespeare, Tom. "The social model of disability." *The disability studies reader 2* (2006): 197-204.

Mismatch between Abilities and Environment⁹

Context-dependent disability results from a mismatch between abilities and the environment:

Ability + Context = Disability

⁹Image source: [Microsoft Inclusive Design Toolkit](#)

Between
humans

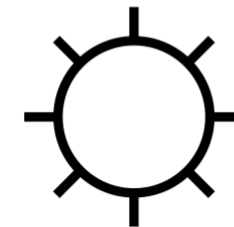


Can't type

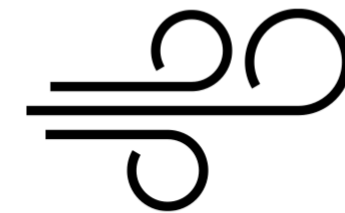


Can't hear

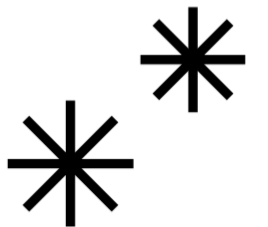
Human+
environment



Glare from sun



Windy



Cold

Human+
object



Left-handed user



Narrow door



Tall shelf

Universal Design¹⁰

Definition: The design of products and environments to be usable by all people, to the greatest extent possible, without the need for adaptation or specialized design.

¹⁰ Ron Mace, 1996

The Main Premise of Universal Design¹¹

Design solutions that benefit some individuals may benefit the whole society. E.g., in the US, only 26K people are suffer loss of upper extremities. Designs that would benefit these 26K would also benefit another 21M people with temporary or situational disabilities.

¹¹ Image source: [Microsoft Inclusive Design Toolkit](#)



An Example: Closed Captioning¹²

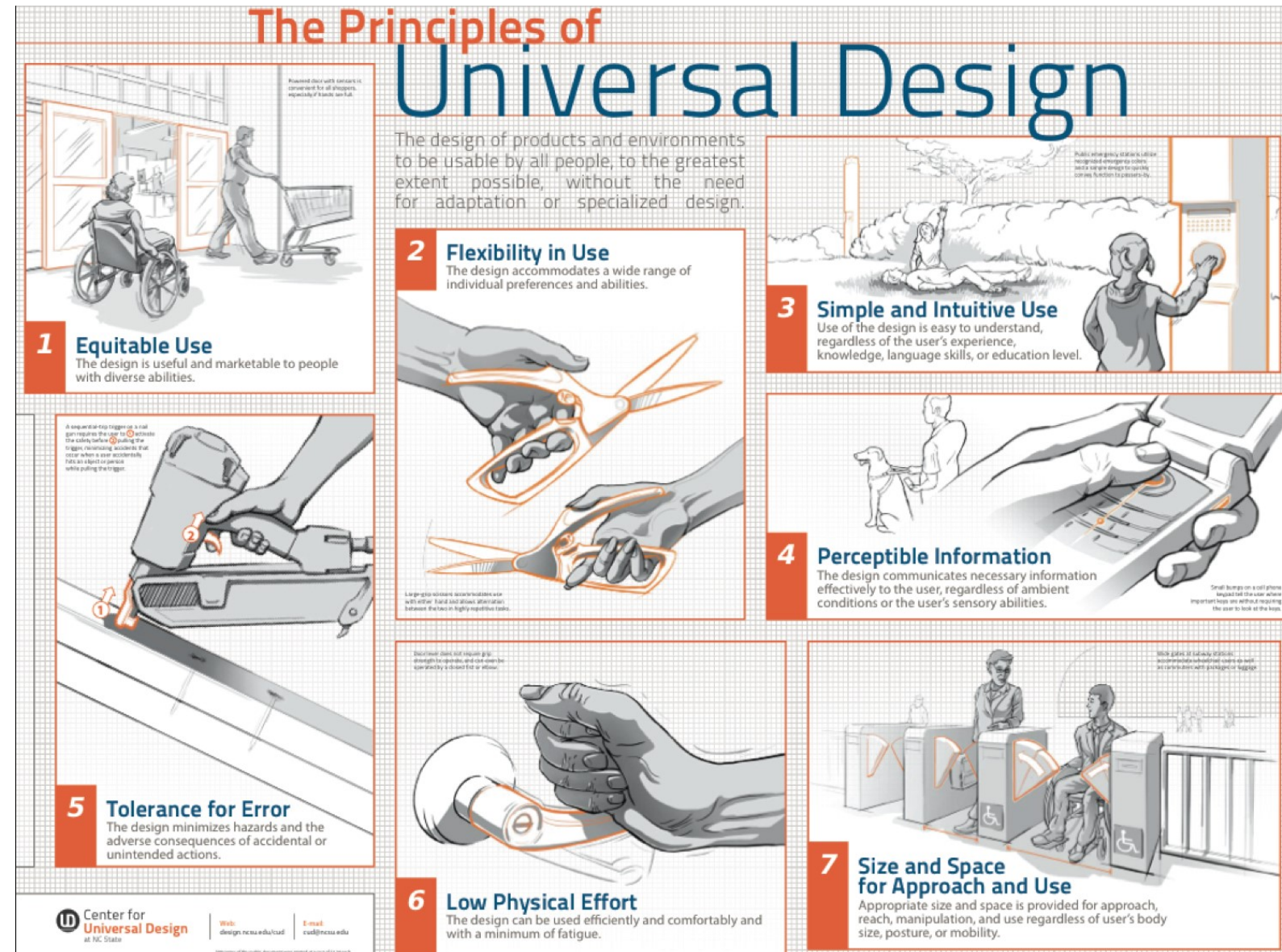
Although closed captioning was originally developed for individuals with hearing impairments, they now also benefit reading in noisy environments and learning to read.

¹² Image source: [Microsoft Inclusive Design Toolkit](#)



Principles of Universal Design

1. Equitable use
2. Flexibility in use
3. Simple and intuitive use
4. Perceptible information
5. Tolerance for error
6. Low physical effort
7. Size and space for approach and use

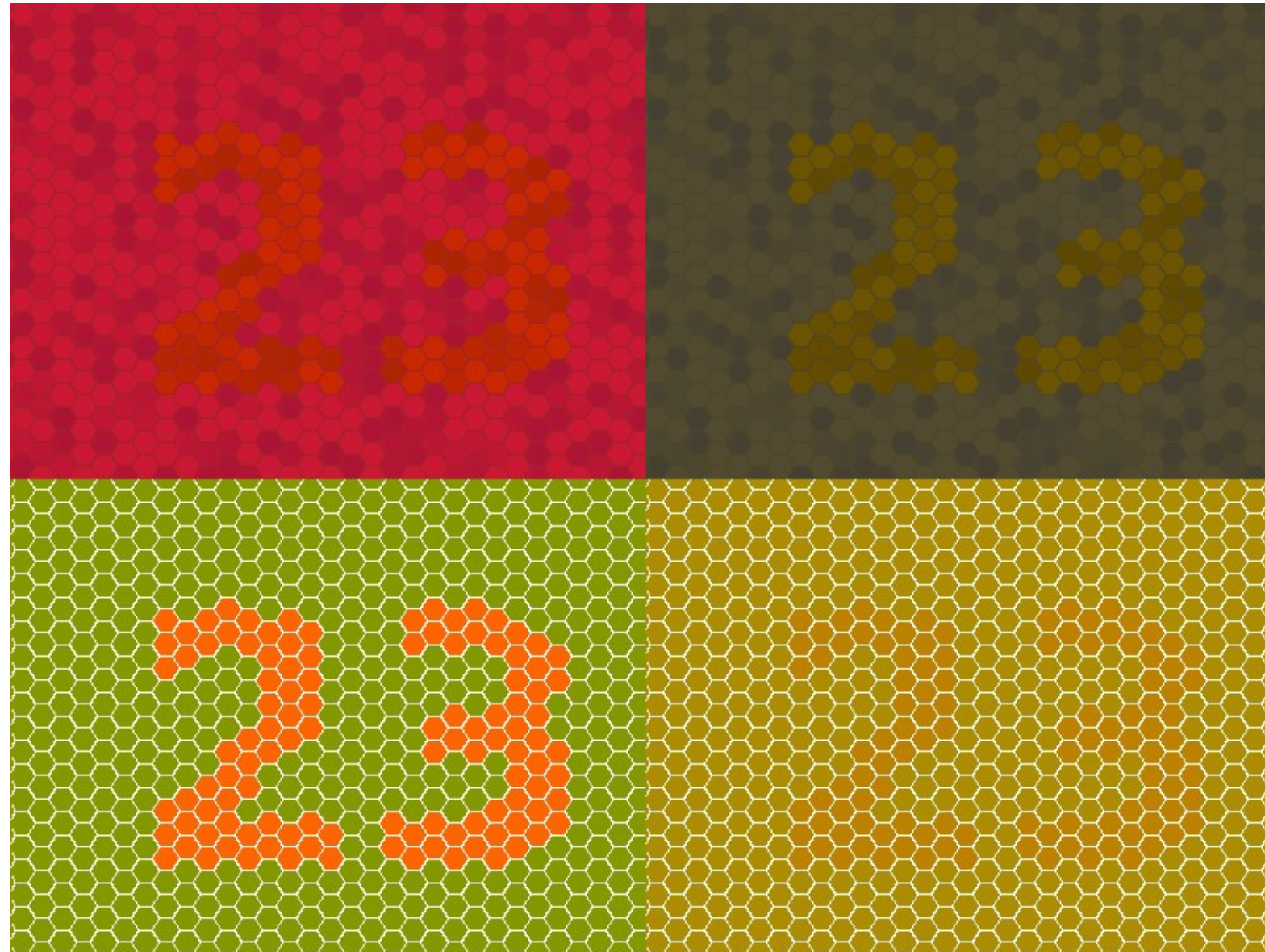


¹³ Image source: Interaction Design Foundation

Principle 1: Equitable use

The design is useful and marketable to people with diverse abilities.

1. Provide the same means of use for all users: identical whenever possible; equivalent when not.
2. Avoid segregating or stigmatizing any users.
3. Provisions for privacy, security, and safety should be equally available to all users.
4. Make the design appealing to all users.



¹⁴ Example source: [Interaction Design Foundation](#); Image source: Johannes Ahlmann

Principle 2: Flexibility in Use

The design accommodates a wide range of individual preferences and abilities.

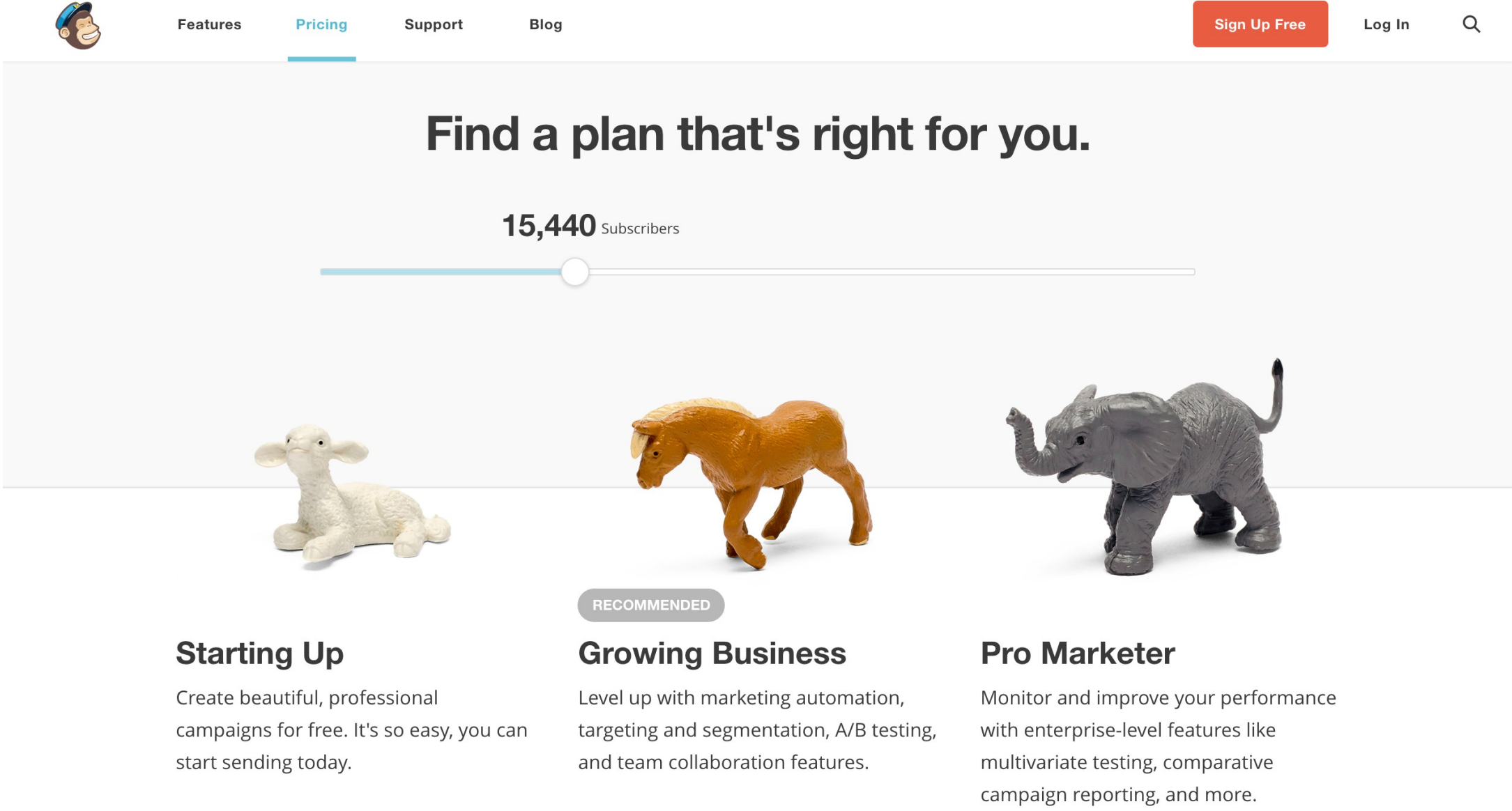
1. Provide choice in methods of use.
2. Accommodate right- or left-handed access and use.
3. Facilitate the user's accuracy and precision.
4. Provide adaptability to the user's pace.

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Principle 3: Simple and Intuitive Use

Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level.

1. Eliminate unnecessary complexity.
2. Be consistent with user expectations and intuition.
3. Accommodate a wide range of literacy and language skills.
4. Arrange information consistent with its importance.
5. Provide effective prompting and feedback during and after task completion.



The screenshot shows a pricing page for a marketing tool. At the top, there is a navigation bar with a logo (a monkey wearing a cap), links for Features, Pricing (highlighted), Support, and Blog. On the right, there are buttons for Sign Up Free, Log In, and a search icon. The main heading is "Find a plan that's right for you." Below this, it says "15,440 Subscribers" with a progress bar. Three pricing plans are displayed, each with a corresponding animal figurine: a white lamb for "Starting Up", an orange horse for "Growing Business" (marked as "RECOMMENDED"), and a grey elephant for "Pro Marketer".

Find a plan that's right for you.

15,440 Subscribers

Starting Up

Create beautiful, professional campaigns for free. It's so easy, you can start sending today.

Growing Business

Level up with marketing automation, targeting and segmentation, A/B testing, and team collaboration features.

Pro Marketer

Monitor and improve your performance with enterprise-level features like multivariate testing, comparative campaign reporting, and more.

¹⁶ Example source: Interaction Design Foundation

Principle 4: Perceptible Information

The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.

1. Use different modes (pictorial, verbal, tactile) for redundant presentation of essential information.
2. Provide adequate contrast between essential information & surroundings.
3. Maximize "legibility" of essential information
4. Differentiate elements in ways that can be described (i.e., make it easy to give instructions or directions).
5. Provide compatibility with a variety of techniques or devices used by people with sensory limitations.



TEDxHouston · Filmed June 2010 · 20:19

Brené Brown: The power of vulnerability

Watch next

Barry Schwartz: Our loss of wisdom

Return to talk

Subtitles and Transcript

Select language

English

0:12

So, I'll start with this: a couple years ago, an event planner called me because I was going to do a speaking event. And she called, and she said, "I'm really struggling with how to write about you on the little flyer." And I thought, "Well, what's the struggle?" And she said, "Well, I saw you speak, and I'm going to call you a researcher, I think, but I'm afraid if I call you a researcher, no one will come, because they'll think you're boring and irrelevant."

0:36

(Laughter)

0:37

And I was like, "Okay." And she said, "But the thing I liked about your talk is you're a storyteller. So I think what I'll do is just call you a storyteller." And of course, the academic, insecure part of me was like, "You're going to call me a what?" And she said, "I'm going to call you a storyteller." And I was like, "Why not 'magic pixie'?"

0:56

(Laughter)

0:59

I was like, "Let me think about this for a second." I tried to call deep on my courage. And I thought, you know, I am a storyteller. I'm a qualitative researcher. I collect stories; that's what I do. And maybe stories are just data with a soul. And maybe I'm just a storyteller. And so I said, "You know what? Why don't you just say I'm a researcher-storyteller." And she went, "Ha ha. There's no such thing."

1:25

(Laughter)

1:27

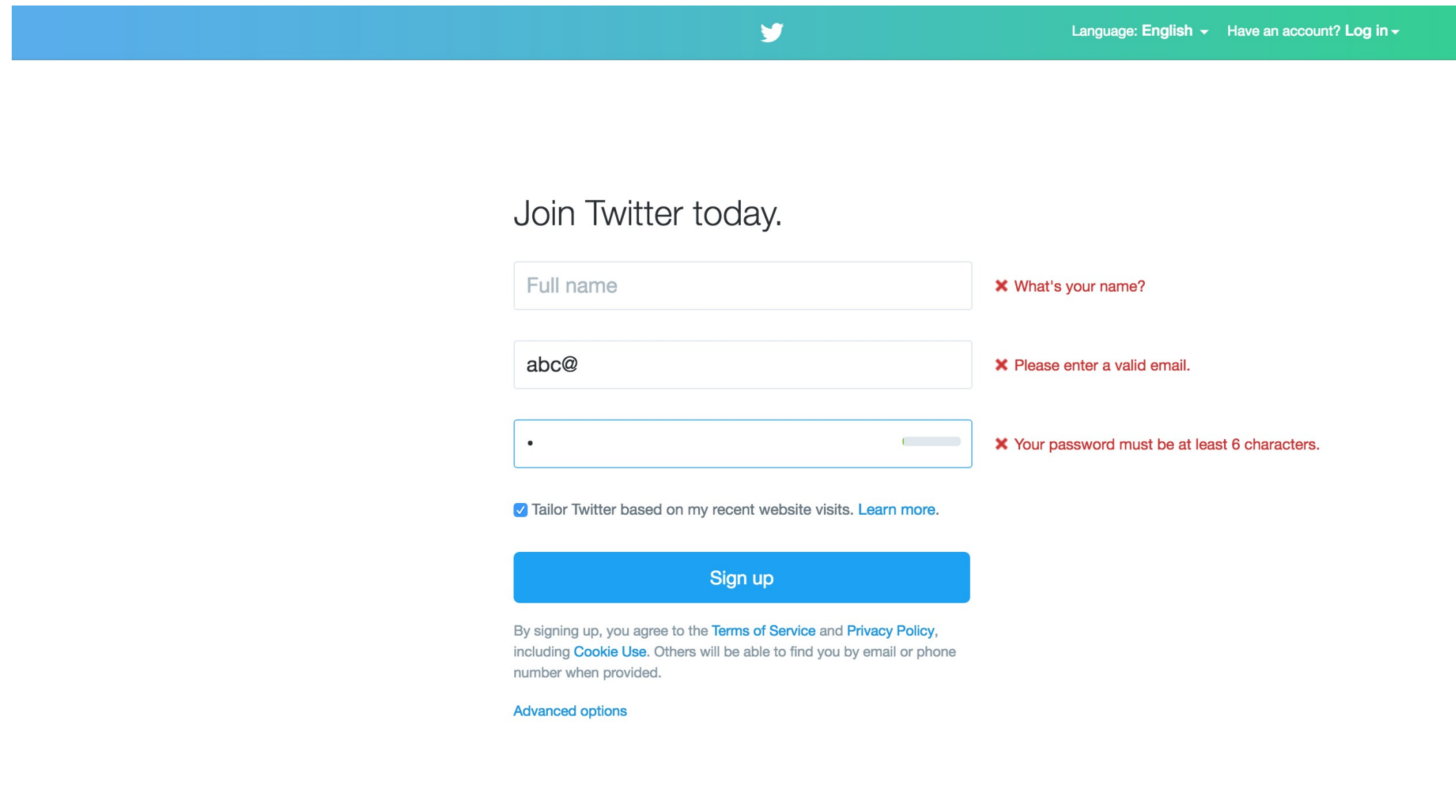
So I'm a researcher-storyteller, and I'm going to talk to you today -- we're talking about expanding perception -- and so I want to talk to you and tell some stories about a piece of my research that fundamentally expanded my perception and really actually changed the way that I live and love and work

¹⁷ Image source: Interaction Design Foundation

Principle 5: Tolerance for Error

The design minimizes hazards and the adverse consequences of accidental or unintended actions.

1. Arrange elements to minimize hazards and errors: most used elements, most accessible; hazardous elements eliminated, isolated, or shielded.
2. Provide warnings of hazards and errors.
3. Provide fail safe features.
4. Discourage unconscious action in tasks that require vigilance.



The image shows a Twitter sign-up form with a blue header bar. The header bar contains the Twitter logo and links for "Language: English" and "Have an account? Log in". The main content area is white and features the text "Join Twitter today." followed by three input fields. The first field is labeled "Full name" and has an error message "What's your name?". The second field contains "abc@" and has an error message "Please enter a valid email.". The third field contains a single dot and has an error message "Your password must be at least 6 characters.". Below the input fields is a checkbox labeled "Tailor Twitter based on my recent website visits. Learn more." and a blue "Sign up" button. At the bottom, there is a paragraph of text about agreeing to the Terms of Service and Privacy Policy, and a link for "Advanced options".

Twitter

Language: English ▾ Have an account? Log in ▾

Join Twitter today.

Full name ✖ What's your name?

abc@ ✖ Please enter a valid email.

• ✖ Your password must be at least 6 characters.

☒ Tailor Twitter based on my recent website visits. [Learn more.](#)

Sign up

By signing up, you agree to the [Terms of Service](#) and [Privacy Policy](#), including [Cookie Use](#). Others will be able to find you by email or phone number when provided.

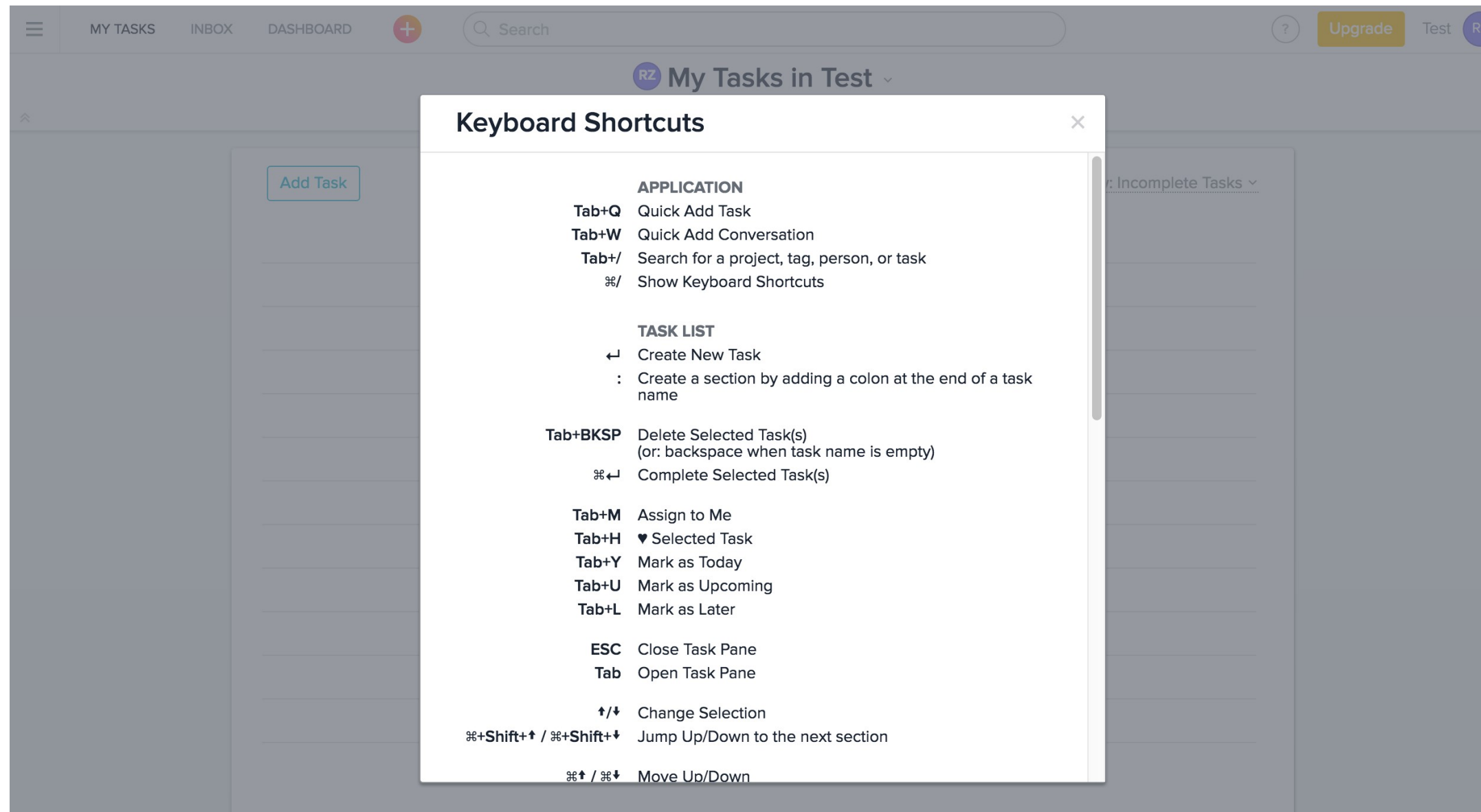
[Advanced options](#)

¹⁸ Image source: [Interaction Design Foundation](#)

Principle 6: Low Physical Effort

The design can be used efficiently and comfortably and with a minimum of fatigue.

1. Allow user to maintain a neutral body position.
2. Use reasonable operating forces.
3. Minimize repetitive actions.
4. Minimize sustained physical effort.



¹⁹ Image source: Interaction Design Foundation

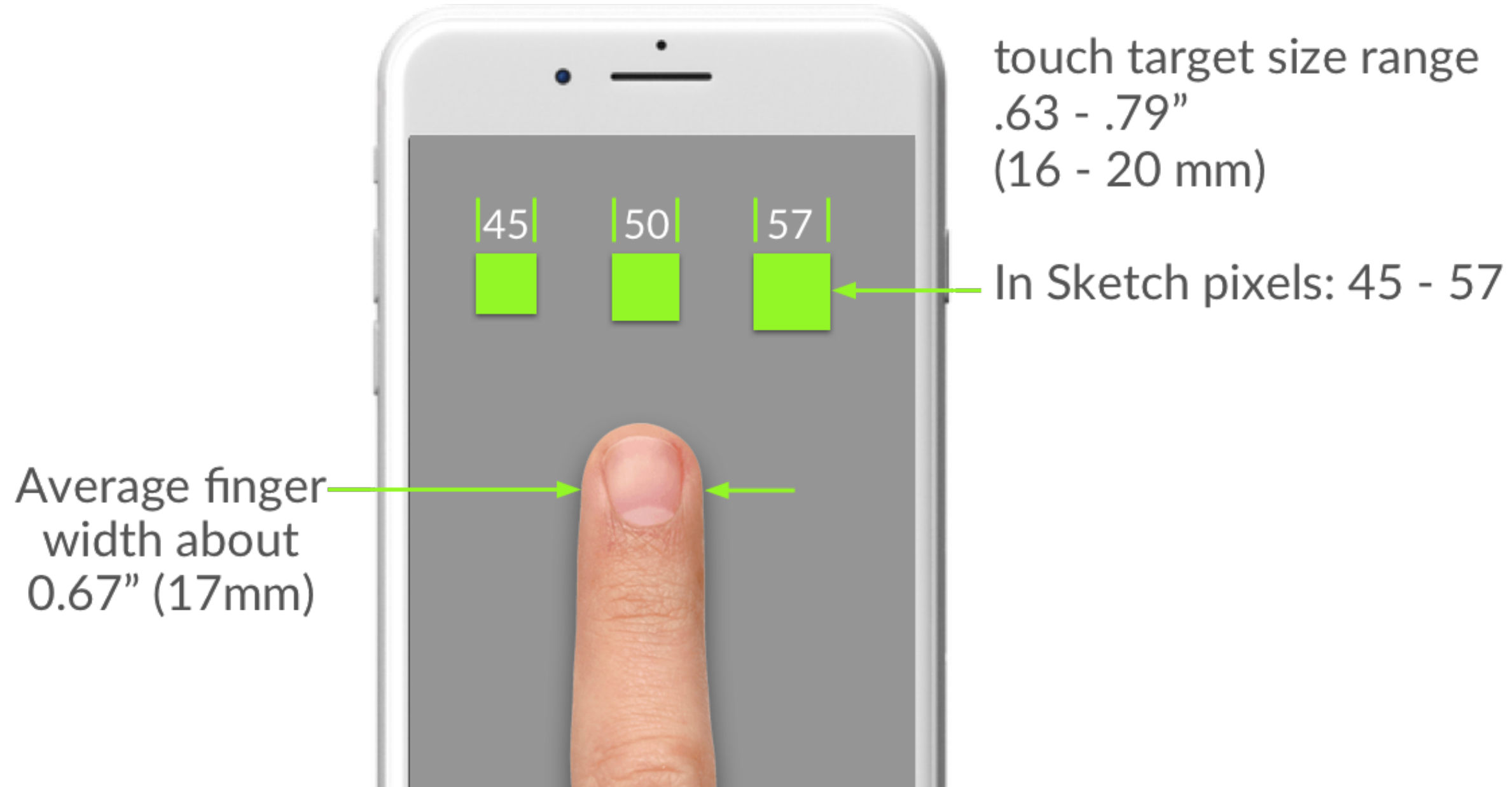
Principle 7: Size and Space for Approach and Use

Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

1. Provide a clear line of sight to important elements for any seated or standing user.
2. Make reach to all components comfortable for any seated or standing user.
3. Accommodate variations in hand and grip size.
4. Provide adequate space for the use of assistive devices or personal assistance.²⁰

²⁰ Image source on next slide

iPhone 8
375 x 667 pixel screen size



Accessible Design for Web Development

Web Content Accessibility Guidelines (WCAG)*

Provides technical requirements for developers and designers.

Three levels of conformance...

1. A (lowest)
2. AA
3. AAA (highest)

Often used as a checklist in evaluation.

*[WCAG Website](#)

Web Content Accessibility Guidelines (WCAG)

All at the A level...

1. Associate a label with every form control
2. Provide a meaningful sequence (i.e. don't skip header levels!)
3. Make errors meaningful and correctable
4. Use sufficient color contrast
5. ... and more!

Remember: This is the floor!

Web Content Accessibility Guidelines (WCAG)

How do we test for conformance?

1. Manual inspection
2. Automated tools
 - a. Wave
 - b. AXE
 - c. ... among many others!

ICA E: Accessibility

Perform an accessibility audit on a website!

Assistive Technologies

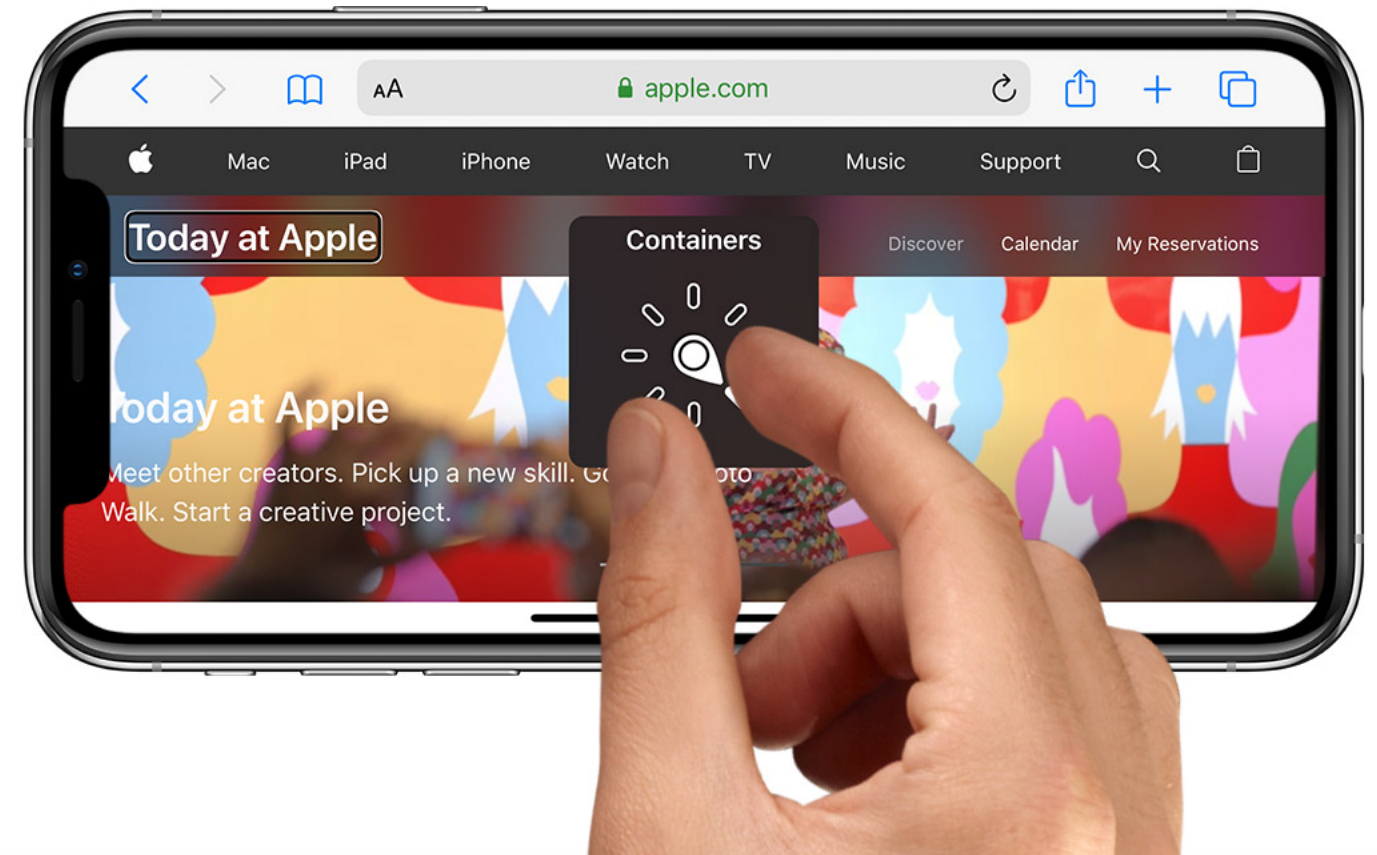
What are assistive technologies?

Definition: Specialized tools that close accessibility gaps.

Screen Readers²¹

Definition: Software used by individuals with vision impairments to read screen content.

- JAWS for Windows
- NVDA
- VoiceOver for MacOS, iOS
- TalkBack for Android



²¹ Image source

Screen Magnification²²

Definition: Enlarges text or graphics on screens to improve visibility of content for individuals with limited vision.

²² Image source



Text Readers²³

Definition: Tools that read out loud text on screens to support vision and learning disabilities.

²³ Image source



Braille for the Web²⁴

Definition: A mechanical device that translates textual content on the screen into Braille.

²⁴ [Image source](#)

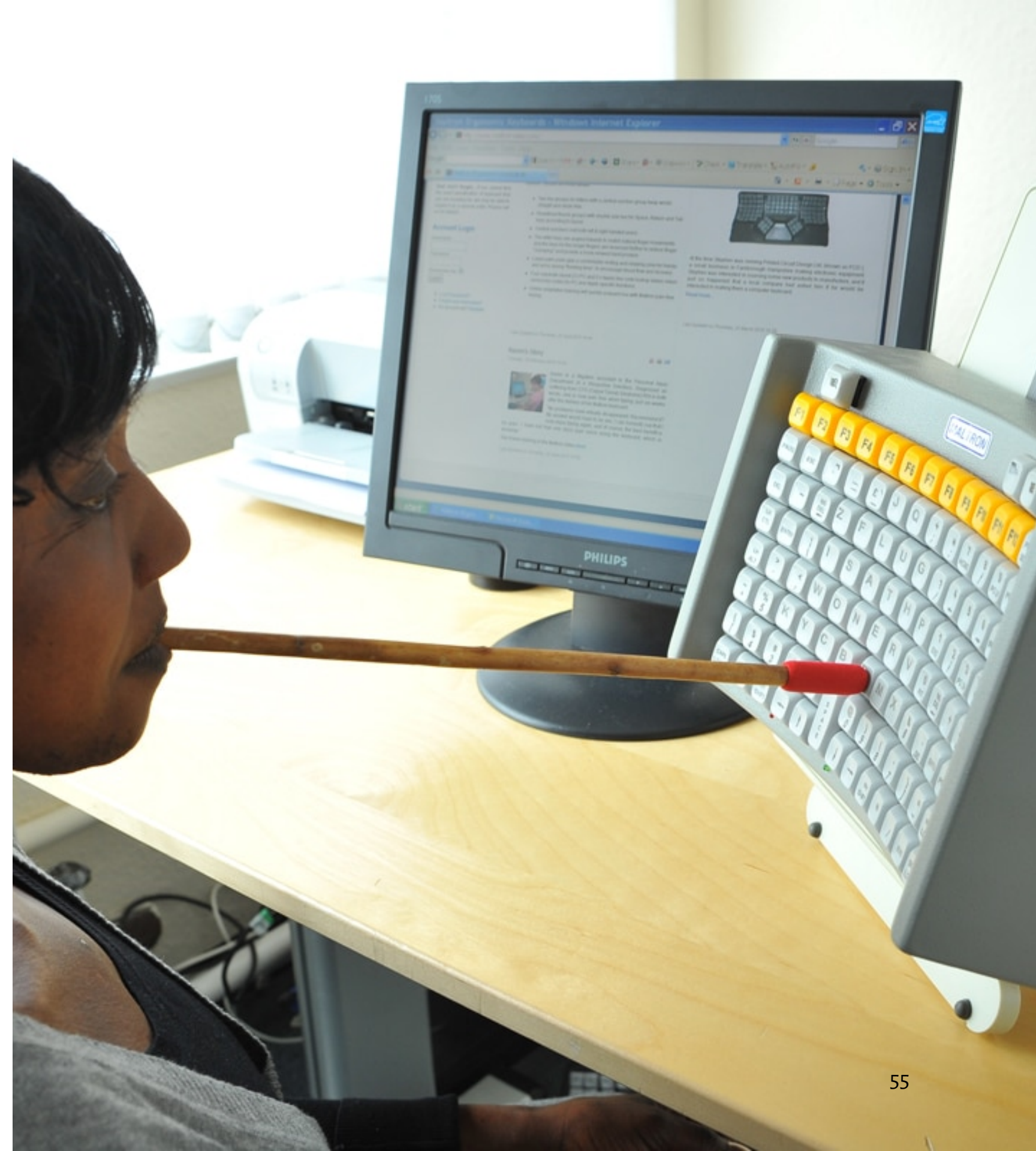


Alternative Input Devices

Definition: Specialized tools that help individuals with motor impairments who cannot use a mouse or keyboard with pointing.

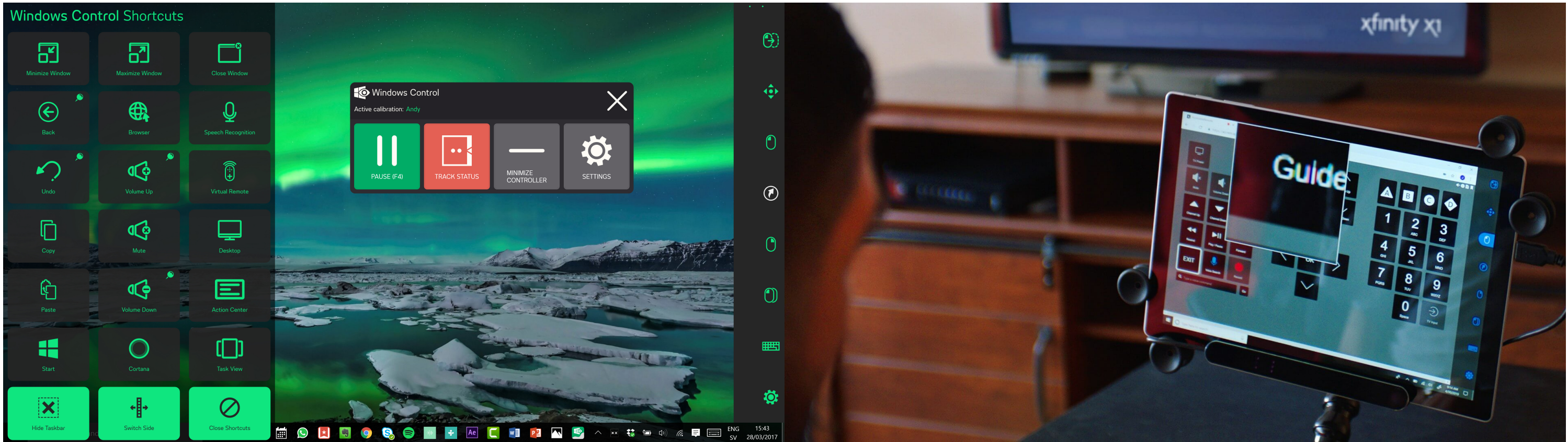
- Head/mouth wands/pointers
- Motion/eye tracking
- Single-switch (e.g., sip-and-puff)
- Speech input

Head/mouth wands/pointers²⁵



²⁵ [Image source](#)

Motion/eye tracking²⁶



²⁶ Image source: left, right

Single-switch (e.g., sip-and-puff)²⁷

²⁷ Image source



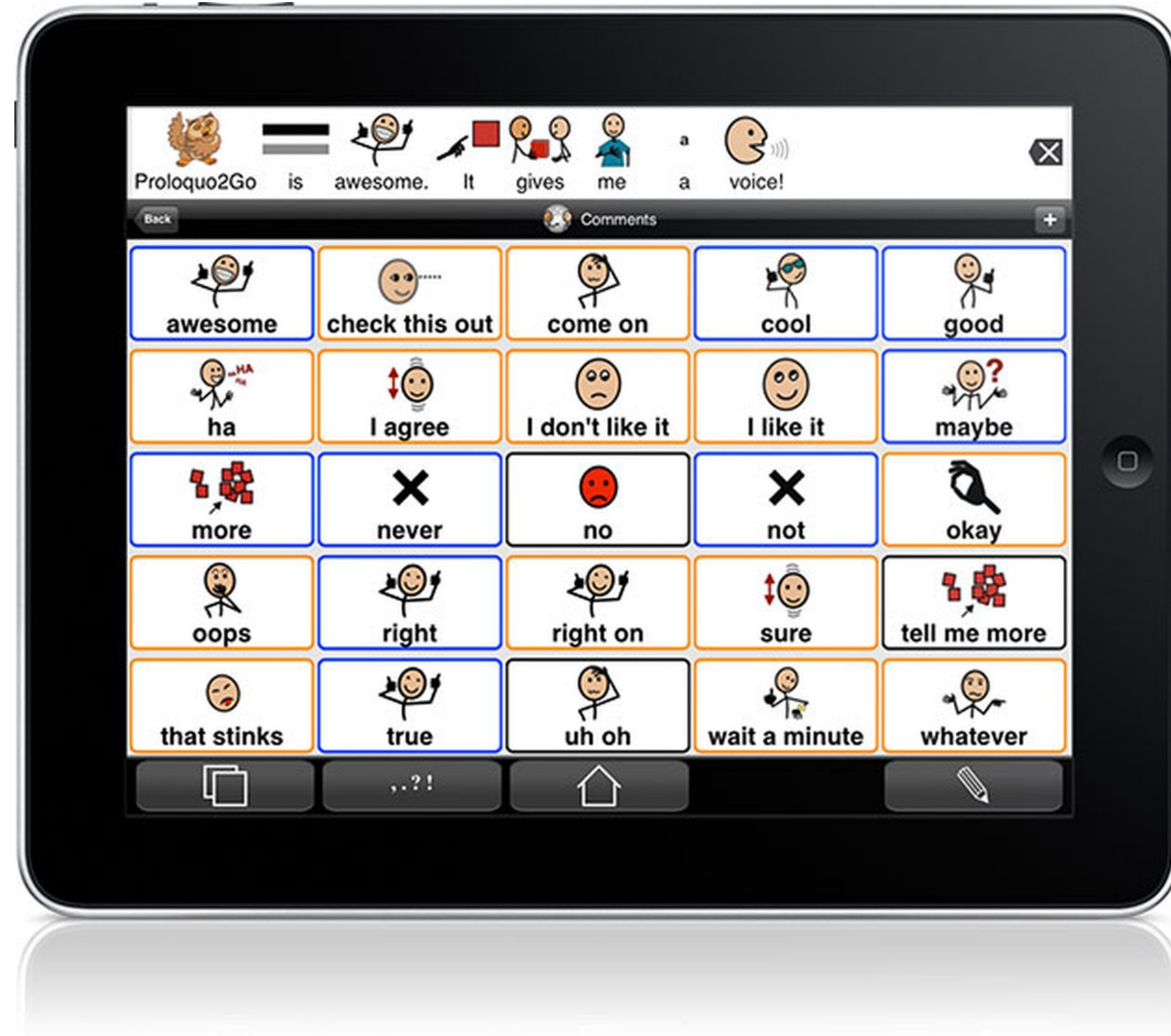
Speech input²⁸

²⁸ Image source



Alternative & Augmentative Communication²⁹

Definition: Tools that help individuals who are unable to use verbal speech to communicate.



²⁹ Image source

ICA E: Accessibility

Use an assistive technology on your mobile device to complete a task!



Navigate your iPhone with VoiceOver

³⁰Video



Android Accessibility

TalkBack

³¹ Video



Questions?